

Moisture Monitor for Plants

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Over-watering and under-watering both are harmful for plants. Roots need air as well as water. If the soil is constantly saturated, air cannot reach the roots and they suffocate. Also, excess water weakens the plant and makes it susceptible to various diseases, particularly fungal attacks. Under-watering, on the other hand, is equally harmful. Plants not receiving enough water droop from the top down and leaf edges turn brown.

Moisture monitor provides a solution to the above problem by monitoring the moisture level of the soil and

producing an audio-visual alert when the moisture goes below a preset level, indicating that the plant needs to be watered. Fig. 1 shows the proposed layout of the device.

Circuit and working

Fig. 2 shows the circuit of moisture monitor for plants.

The core of the circuit is inexpensive and built around popular 14-stage ripple-carry binary counter CD4060 (IC1) and a few commonly available components. Not all the outputs of the binary counter are available externally. The frequency at pin 7 (Q3) of IC1 is actually divided by $2^4=16$ compared to initial astable frequency. Similarly,

Q4 will have half the frequency of Q3. Components R3, R4 and C2 set the clock oscillator frequency to control the flash rate of LED1 and the beeping rate of PZ1. Pulses are enabled when reset pin (pin 12) is low and disabled, or inhibited, when reset pin is high.

Working of the circuit is simple and straightforward. The base of trans-

Test Points

Test point	Details
TP0	0 V, GND
TP1	9 V when switch S1 is on
TP2	High (8 V approx.) when soil is wet and low when soil is dry
TP3	Pulsetrain when soil is dry

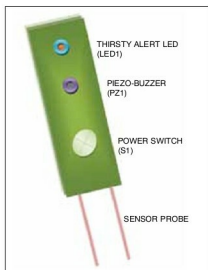


Fig. 1: Author's prototype

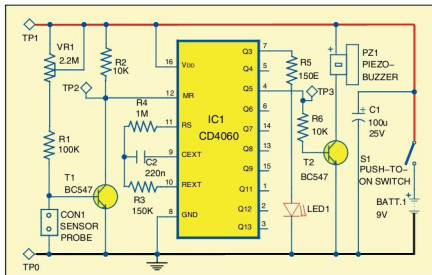


Fig. 2: Circuit of moisture monitor for plants

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